1、 启动 vLLM

```
(mllm124) [root:zzz]$ vllm serve Qwen/Qwen2.5-VL-3B-Instruct   --host 0.0.0.0   --port 8000   --limit-mm-per-prompt '{"image":1,"video":0}'   --max-model-len 32768
 INFO 05-04 15:28:40 [__init__.py:239] Automatically detected platform cuda.
 INFO 05-04 15:28:45 [api_server.py:1043] vLLM API server version 0.8.5.post1
 INFO 05-04 15:28:45 [api_server.py:1044] args: Namespace(subparser='serve', model_tag='Qwen/Qwen2.5-VL-3B-Instruct', config='', host='0.0.0.0', port=8000, uvicorn_log_leve
```

```
INFO 05-04 15:29:14 [weight_utils.py:265] Using model weights format ['*.safetensors']
Loading safetensors checkpoint shards:   0% Completed | 0/2 [00:00<?, ?it/s]
Loading safetensors checkpoint shards:  50% Completed | 1/2 [00:00<00:00,  1.05it/s]
Loading safetensors checkpoint shards: 100% Completed | 2/2 [00:02<00:00,  1.07s/it]
Loading safetensors checkpoint shards: 100% Completed | 2/2 [00:02<00:00,  1.05s/it]

INFO 05-04 15:29:17 [loader.py:458] Loading weights took 2.41 seconds
INFO 05-04 15:29:17 [gpu_model_runner.py:1347] Model loading took 7.1557 GiB and 4.181624 seconds
INFO 05-04 15:29:18 [gpu_model_runner.py:1620] Encoder cache will be initialized with a budget of 16384 tokens, and profiled with 1 image items of the maximum feature siz
.
INFO 05-04 15:29:32 [backends.py:420] Using cache directory: /root/.cache/vllm/torch_compile_cache/98b22d218c/rank_0_0 for vLLM's torch.compile
INFO 05-04 15:29:32 [backends.py:430] Dynamo bytecode transform time: 8.66 s
INFO 05-04 15:29:37 [backends.py:118] Directly load the compiled graph(s) for shape None from the cache, took 5.031 s
INFO 05-04 15:29:39 [monitor.py:33] torch.compile takes 8.66 s in total
INFO 05-04 15:29:39 [kv_cache_utils.py:634] GPU KV cache size: 898,944 tokens
INFO 05-04 15:29:39 [kv_cache_utils.py:637] Maximum concurrency for 32,768 tokens per request: 27.43x
INFO 05-04 15:30:00 [gpu_model_runner.py:1686] Graph capturing finished in 21 secs, took 0.57 GiB
INFO 05-04 15:30:00 [core.py:159] init engine (profile, create kv cache, warmup model) took 43.06 seconds
INFO 05-04 15:30:01 [core_client.py:439] Core engine process 0 ready.
WARNING 05-04 15:30:02 [config.py:1239] Default sampling parameters have been overridden by the model's Hugging Face generation config recommended from the model creator.
If this is not intended, please relaunch vLLM instance with `--generation-config vllm`.
INFO 05-04 15:30:02 [serving_chat.py:118] Using default chat sampling params from model: {'repetition_penalty': 1.05, 'temperature': 1e-06}
INFO 05-04 15:30:02 [serving_completion.py:61] Using default completion sampling params from model: {'repetition_penalty': 1.05, 'temperature': 1e-06}
INFO 05-04 15:30:02 [api_server.py:1090] Starting vLLM API server on http://0.0.0.0:8000
INFO 05-04 15:30:02 [launcher.py:28] Available routes are:
```

```
INFO 05-04 15:30:02 [launcher.py:28] Available routes are:
INFO 05-04 15:30:02 [launcher.py:36] Route: /openapi.json, Methods: GET, HEAD
INFO 05-04 15:30:02 [launcher.py:36] Route: /docs, Methods: GET, HEAD
INFO 05-04 15:30:02 [launcher.py:36] Route: /docs/oauth2-redirect, Methods: GET, HEAD
INFO 05-04 15:30:02 [launcher.py:36] Route: /redoc, Methods: GET, HEAD
INFO 05-04 15:30:02 [launcher.py:36] Route: /health, Methods: GET
INFO 05-04 15:30:02 [launcher.py:36] Route: /load, Methods: GET
INFO 05-04 15:30:02 [launcher.py:36] Route: /ping, Methods: POST, GET
INFO 05-04 15:30:02 [launcher.py:36] Route: /tokenize, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /detokenize, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /v1/models, Methods: GET
INFO 05-04 15:30:02 [launcher.py:36] Route: /version, Methods: GET
INFO 05-04 15:30:02 [launcher.py:36] Route: /v1/chat/completions, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /v1/completions, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /v1/embeddings, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /pooling, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /score, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /v1/score, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /v1/audio/transcriptions, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /rerank, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /v1/rerank, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /v2/rerank, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /invocations, Methods: POST
INFO 05-04 15:30:02 [launcher.py:36] Route: /metrics, Methods: GET
INFO:     Started server process [1380764]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

2、数据准备检查

```
(mllm124) [root:zzz]$ wc -l data/processed/mathvision_testmini.jsonl data/processed/vstar_bench.jsonl
   304 data/processed/mathvision_testmini.jsonl
   191 data/processed/vstar_bench.jsonl
   495 total
```

3、MathVision Baseline

```
(mllm124) [root:zzz]$
  python scripts/run_inference.py \
    --input data/processed/mathvision_testmini.jsonl \
    --output outputs/predictions/mathvision_testmini_predictions.jsonl \
    --batch-size 1
Running inference:  96%|
                         | 291/304 [38:04<01:12,  5.55s/batch, s
Running inference:  96%|
                        | 292/304 [38:04<43:42, 218.52s/batch, s
Running inference:  96%|
                        | 292/304 [38:11<43:42, 218.52s/batch, s
Running inference:  96%|
                        | 293/304 [38:11<28:27, 155.18s/batch, s
Running inference:  96%|
                        | 293/304 [38:20<28:27, 155.18s/batch, s
Running inference:  97%|
                        | 294/304 [38:20<18:32, 111.25s/batch, s
Running inference:  97%|
                        | 294/304 [38:23<18:32, 111.25s/batch, s
Running inference:  97%|
                         | 295/304 [38:23<11:49, 78.88s/batch, s
Running inference:  97%|
                         | 295/304 [38:28<11:49, 78.88s/batch, s
Running inference:  97%|
                         | 296/304 [38:28<07:33, 56.69s/batch, s
Running inference:  97%|
                         | 296/304 [38:34<07:33, 56.69s/batch, s
Running inference:  98%|
                         | 297/304 [38:34<04:50, 41.46s/batch, s
Running inference: 100%|                    | 304/304 [39:30<00:00,  7.80s/batch, samples=304]
Saved predictions -> outputs/predictions/mathvision_testmini_predictions.jsonl
```

（推理↑ 评测↓）

```
(mllm124) [root:zzz]$   python scripts/run_eval.py \
    --predictions outputs/predictions/mathvision_testmini_predictions.jsonl \
    --report outputs/eval_reports/mathvision_testmini_report.json \
    --bad-cases outputs/eval_reports/mathvision_testmini_bad_cases.jsonl
Evaluating predictions: 100%|| 304/304 [00:00<00:00, 22927.11sample/s]
{'total': 304, 'correct': 38, 'accuracy': 0.125, 'parsed': 304, 'parse_success_rate': 1.0, 'by_dataset': {'mathvision': {'total': 304, 'correct': 38, 'accuracy': 0.125}},
'predictions_path': 'outputs/predictions/mathvision_testmini_predictions.jsonl', 'bad_cases_path': 'outputs/eval_reports/mathvision_testmini_bad_cases.jsonl'}
```

4、VBench Baseline 对应内容

```
(mllm124) [root:zzz]$   python scripts/run_inference.py \
    --input data/processed/vstar_bench.jsonl \
    --output outputs/predictions/vstar_bench_predictions.jsonl \
    --batch-size 1
Running inference: 100%|| 191/191 [10:13<00:00,  3.21s/batch, samples=191]
Saved predictions -> outputs/predictions/vstar_bench_predictions.jsonl
```

（推理↑ 评测↓）

```
(mllm124) [root:zzz]$   python scripts/run_eval.py \
    --predictions outputs/predictions/vstar_bench_predictions.jsonl \
    --report outputs/eval_reports/vstar_bench_report.json \
    --bad-cases outputs/eval_reports/vstar_bench_bad_cases.jsonl
Evaluating predictions: 100%|| 191/191 [00:00<00:00, 88354.70sample/s]
{'total': 191, 'correct': 143, 'accuracy': 0.7486910994764397, 'parsed': 191, 'parse_success_rate': 1.0, 'by_dataset': {'vstar_bench': {'total': 191, 'correct': 143, 'accu
racy': 0.7486910994764397}}, 'predictions_path': 'outputs/predictions/vstar_bench_predictions.jsonl', 'bad_cases_path': 'outputs/eval_reports/vstar_bench_bad_cases.jsonl'}
```

5、MathVision Self-consistency N=4

```
(mllm124) [root:zzz]$ python scripts/run_self_consistency.py \
    --input data/processed/mathvision_testmini.jsonl \
    --output outputs/predictions/mathvision_self_consistency_n4_predictions.jsonl \
    --candidates-output outputs/predictions/mathvision_self_consistency_n4_candidates.jsonl \
    --n 4 \
    --batch-size 1
Self-consistency N=4:   6%|
                          | 79/1216 [18:01<69:54:31, 221.35s/batch, candidate=1,
Self-consistency N=4:   6%|
                          | 79/1216 [18:01<69:54:31, 221.35s/batch, candidate=1,
Self-consistency N=4:   7%|
                          | 80/1216 [18:05<49:16:44, 156.17s/batch, candidate=1,
Self-consistency N=4:   7%|
                          | 80/1216 [18:05<49:16:44, 156.17s/batch, candidate=1,
Self-consistency N=4:   7%|
                          | 81/1216 [18:13<35:13:18, 111.72s/batch, candidate=1,
Self-consistency N=4:   7%|
                          | 81/1216 [18:13<35:13:18, 111.72s/batch, candidate=1,
Self-consistency N=4:   9%|          | 104/1216 [21:28<2:22:33,  7.69s/batch, candidate=1, samples=104]
```

（推理↑ 评测↓）

```
(mllm124) [root:zzz]$
  python scripts/run_eval.py \
    --predictions outputs/predictions/mathvision_self_consistency_n4_predictions.jsonl \
    --report outputs/eval_reports/mathvision_self_consistency_n4_report.json \
    --bad-cases outputs/eval_reports/mathvision_self_consistency_n4_bad_cases.jsonl
Evaluating predictions: 100%|| 304/304 [00:00<00:00, 23014.43sample/s]
{'total': 304, 'correct': 65, 'accuracy': 0.2138157894736842, 'parsed': 304, 'parse_success_rate': 1.0, 'by_dataset': {'mathvision': {'total': 304, 'correct': 65, 'accurac
y': 0.2138157894736842}}, 'predictions_path': 'outputs/predictions/mathvision_self_consistency_n4_predictions.jsonl', 'bad_cases_path': 'outputs/eval_reports/mathvision_se
lf_consistency_n4_bad_cases.jsonl'}
```

6、VBench Self-consistency N=4 对应内容

```
(mllm124) [root:zzz]$
  python scripts/run_self_consistency.py \
    --input data/processed/vstar_bench.jsonl \
    --output outputs/predictions/vstar_self_consistency_n4_predictions.jsonl \
    --candidates-output outputs/predictions/vstar_self_consistency_n4_candidates.jsonl \
    --n 4 \
    --batch-size 1
Self-consistency N=4:  23%|          | 177/764 [09:19<34:43,  3.55s/batch, candidate=1, sSelf-consis
```

（推理↑ 评测↓）

```
(mllm124) [root:zzz]$   python scripts/run_eval.py \
    --predictions outputs/predictions/vstar_self_consistency_n4_predictions.jsonl \
    --report outputs/eval_reports/vstar_self_consistency_n4_report.json \
    --bad-cases outputs/eval_reports/vstar_self_consistency_n4_bad_cases.jsonl
Evaluating predictions: 100%|| 191/191 [00:00<00:00, 66826.16sample/s]
{'total': 191, 'correct': 137, 'accuracy': 0.7172774869109948, 'parsed': 191, 'parse_success_rate': 1.0, 'by_dataset': {'vstar_bench': {'total': 191, 'correct': 137, 'accu
racy': 0.7172774869109948}}, 'predictions_path': 'outputs/predictions/vstar_self_consistency_n4_predictions.jsonl', 'bad_cases_path': 'outputs/eval_reports/vstar_self_cons
istency_n4_bad_cases.jsonl'}
```

7、Oracle@4 分析

```
(mllm124) [root:zzz]$   python scripts/analyze_candidates.py \
    --candidates outputs/predictions/mathvision_self_consistency_n4_candidates.jsonl \
    --report outputs/eval_reports/mathvision_self_consistency_n4_candidates_report.json \
    --details-output outputs/eval_reports/mathvision_self_consistency_n4_candidates_details.jsonl
Analyzing candidates: 100%|| 304/304 [00:00<00:00, 4398.53sample/s]
{'total': 304, 'num_candidates_per_sample': [4], 'oracle_correct': 121, 'oracle_at_n': 0.3980263157894737, 'majority_correct': 65, 'majority_accuracy': 0.2138157894736842,
 'oracle_majority_gap': 0.18421052631578946, 'oracle_not_majority': 56, 'diverse_answer_samples': 296, 'diverse_answer_rate': 0.9736842105263158, 'avg_unique_answers': 3.2
532894736842106, 'avg_consensus_rate': 0.42680921052631576}
(mllm124) [root:zzz]$   python scripts/analyze_candidates.py \
    --candidates outputs/predictions/vstar_self_consistency_n4_candidates.jsonl \
    --report outputs/eval_reports/vstar_self_consistency_n4_candidates_report.json \
    --details-output outputs/eval_reports/vstar_self_consistency_n4_candidates_details.jsonl
Analyzing candidates: 100%|| 191/191 [00:00<00:00, 15775.51sample/s]
{'total': 191, 'num_candidates_per_sample': [4], 'oracle_correct': 167, 'oracle_at_n': 0.8743455497382199, 'majority_correct': 137, 'majority_accuracy': 0.7172774869109948
, 'oracle_majority_gap': 0.15706806282722513, 'oracle_not_majority': 30, 'diverse_answer_samples': 120, 'diverse_answer_rate': 0.6282722513089005, 'avg_unique_answers': 1.
8324607329842932, 'avg_consensus_rate': 0.768324607329843}
```

8、Verifier / Reranker 训练

```
(mllm124) [root:zzz]$   python scripts/rule_verifier_rerank.py \
    --candidates outputs/predictions/mathvision_self_consistency_n4_candidates.jsonl \
    --output outputs/predictions/mathvision_rule_verifier_n4_predictions.jsonl \
    --scores-output outputs/eval_reports/mathvision_rule_verifier_n4_scores.jsonl
Rule verifier rerank: 100%|| 304/304 [00:00<00:00, 1656.27sample/s]
Saved rule-verifier predictions -> outputs/predictions/mathvision_rule_verifier_n4_predictions.jsonl
Saved rule-verifier scores -> outputs/eval_reports/mathvision_rule_verifier_n4_scores.jsonl
(mllm124) [root:zzz]$   python scripts/run_eval.py \
    --predictions outputs/predictions/mathvision_rule_verifier_n4_predictions.jsonl \
    --report outputs/eval_reports/mathvision_rule_verifier_n4_report.json \
    --bad-cases outputs/eval_reports/mathvision_rule_verifier_n4_bad_cases.jsonl
Evaluating predictions: 100%|| 304/304 [00:00<00:00, 27380.20sample/s]
{'total': 304, 'correct': 64, 'accuracy': 0.21052631578947367, 'parsed': 304, 'parse_success_rate': 1.0, 'by_dataset': {'mathvision': {'total': 304, 'correct': 64, 'accura
cy': 0.21052631578947367}}, 'predictions_path': 'outputs/predictions/mathvision_rule_verifier_n4_predictions.jsonl', 'bad_cases_path': 'outputs/eval_reports/mathvision_rul
e_verifier_n4_bad_cases.jsonl'}
```

```
(mllm124) [root:zzz]$   python scripts/train_verifier.py \
    --config configs/verifier_mathvision_n4.yaml
{'train': 976, 'valid': 240, 'train_positive': 138, 'valid_positive': 47, 'class_weights': [0.5823389291763306, 3.5362319946289062]}
Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized: ['classifier.bias'
, 'classifier.weight', 'pre_classifier.bias', 'pre_classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Map: 100%|| 976/976 [00:00<00:00, 3390.83 examples/s]
Map: 100%|| 240/240 [00:00<00:00, 3203.13 examples/s]
/inspire/hdd/project/powersystem/fengkairui-25026/wangjiayi/zzz/scripts/train_verifier.py:27: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0
 for `WeightedTrainer.__init__`. Use `processing_class` instead.
  super().__init__(*args, **kwargs)
{'loss': 0.6954, 'grad_norm': 2.067847967147827, 'learning_rate': 1.7923497267759563e-05, 'epoch': 0.33}
{'loss': 0.6819, 'grad_norm': 2.5750632286071777, 'learning_rate': 1.5737704918032788e-05, 'epoch': 0.66}
{'loss': 0.6916, 'grad_norm': 2.1967368125915527, 'learning_rate': 1.3551912568306011e-05, 'epoch': 0.98}
{'eval_loss': 0.6980336308479309, 'eval_accuracy': 0.7416666666666667, 'eval_runtime': 0.8423, 'eval_samples_per_second': 284.922, 'eval_steps_per_second': 17.808, 'epoch'
: 1.0}
{'loss': 0.6815, 'grad_norm': 3.3461787700653076, 'learning_rate': 1.1366120218579235e-05, 'epoch': 1.31}
{'loss': 0.6805, 'grad_norm': 3.319347381591797, 'learning_rate': 9.18032786885246e-06, 'epoch': 1.64}
{'loss': 0.6462, 'grad_norm': 3.2961132526397705, 'learning_rate': 6.994535519125684e-06, 'epoch': 1.97}
{'eval_loss': 0.8584661483764648, 'eval_accuracy': 0.8041666666666667, 'eval_runtime': 0.8295, 'eval_samples_per_second': 289.328, 'eval_steps_per_second': 18.083, 'epoch'
: 2.0}
{'loss': 0.6, 'grad_norm': 6.294055461883545, 'learning_rate': 4.808743169398907e-06, 'epoch': 2.3}
{'loss': 0.6298, 'grad_norm': 3.821157693862915, 'learning_rate': 2.6229508196721314e-06, 'epoch': 2.62}
{'loss': 0.6099, 'grad_norm': 6.071598529815674, 'learning_rate': 4.3715846994535524e-07, 'epoch': 2.95}
{'eval_loss': 0.7870451211929321, 'eval_accuracy': 0.625, 'eval_runtime': 0.8551, 'eval_samples_per_second': 280.676, 'eval_steps_per_second': 17.542, 'epoch': 3.0}
{'train_runtime': 37.3713, 'train_samples_per_second': 78.349, 'train_steps_per_second': 4.897, 'train_loss': 0.6554494073482159, 'epoch': 3.0}
100%|| 183/183 [00:37<00:00,  4.90it/s]
```

```
(mllm124) [root:zzz]$   python scripts/score_with_verifier.py \
    --candidates outputs/predictions/mathvision_self_consistency_n4_candidates.jsonl \
    --verifier-dir outputs/verifier/mathvision_n4_checkpoints \
    --output outputs/predictions/mathvision_trained_verifier_n4_predictions.jsonl \
    --report outputs/eval_reports/mathvision_trained_verifier_n4_report.json \
    --scores-output outputs/eval_reports/mathvision_trained_verifier_n4_scores.jsonl \
    --report-subset-ids-from outputs/verifier/mathvision_n4_valid.jsonl \
    --batch-size 16
Parsing candidate answers: 100%|| 1216/1216 [00:00<00:00, 2037664.27candidate/s]
Scoring candidates: 100%|| 76/76 [00:02<00:00, 27.99batch/s]
Grouping candidates: 100%|| 1216/1216 [00:00<00:00, 1212042.22candidate/s]
Selecting best candidates: 100%|| 304/304 [00:00<00:00, 18919.61question/s]
{'total': 60, 'correct': 15, 'accuracy': 0.25, 'parsed': 60, 'parse_success_rate': 1.0, 'by_dataset': {'mathvision': {'total': 60, 'correct': 15, 'accuracy': 0.25}}, 'repo
rt_subset_ids_from': 'outputs/verifier/mathvision_n4_valid.jsonl', 'selected_total_before_subset': 304}
(mllm124) [root:zzz]$   python scripts/mllm_verifier_rerank.py \
    --model-config configs/model.yaml \
    --candidates outputs/predictions/mathvision_self_consistency_n4_candidates.jsonl \
    --output outputs/predictions/mathvision_mllm_verifier_n4_predictions.jsonl \
    --report outputs/eval_reports/mathvision_mllm_verifier_n4_report.json \
    --judge-output outputs/eval_reports/mathvision_mllm_verifier_n4_judge.jsonl
MLLM verifier rerank: 100%|| 304/304 [03:06<00:00,  1.63sample/s]
{'total': 304, 'correct': 53, 'accuracy': 0.17434210526315788, 'parsed': 304, 'parse_success_rate': 1.0, 'by_dataset': {'mathvision': {'total': 304, 'correct': 53, 'accura
cy': 0.17434210526315788}}, 'candidates_path': 'outputs/predictions/mathvision_self_consistency_n4_candidates.jsonl', 'model': 'Qwen/Qwen2.5-VL-3B-Instruct'}
```

9、Bad Case 分析

```
(mllm124) [root:zzz]$   python scripts/summarize_bad_cases.py \
    --predictions outputs/predictions/mathvision_self_consistency_n4_predictions.jsonl \
    --candidates outputs/predictions/mathvision_self_consistency_n4_candidates.jsonl \
    --report outputs/eval_reports/mathvision_bad_case_summary.json \
    --samples-output outputs/eval_reports/mathvision_bad_case_samples.jsonl \
    --max-samples-per-category 8
Summarizing bad cases: 100%|| 239/239 [00:00<00:00, 4976.88case/s]
{'predictions_path': 'outputs/predictions/mathvision_self_consistency_n4_predictions.jsonl', 'candidates_path': 'outputs/predictions/mathvision_self_consistency_n4_candida
tes.jsonl', 'total_predictions': 304, 'bad_cases': 239, 'bad_case_rate': 0.7861842105263158, 'category_counts': {'candidate_generation_failure': 183, 'reranker_or_voting_f
ailure': 56}, 'problem_type_counts': {'counting': 73, 'open_visual_reasoning': 9, 'multiple_choice_visual_reasoning': 46, 'geometry_or_measurement': 66, 'chart_or_table':
20, 'ocr_or_text_reading': 25}, 'sampled_cases': 16}
(mllm124) [root:zzz]$   python scripts/summarize_bad_cases.py \
    --predictions outputs/predictions/vstar_bench_predictions.jsonl \
    --candidates outputs/predictions/vstar_self_consistency_n4_candidates.jsonl \
    --report outputs/eval_reports/vstar_bad_case_summary.json \
    --samples-output outputs/eval_reports/vstar_bad_case_samples.jsonl \
    --max-samples-per-category 8
Summarizing bad cases: 100%|| 191/191 [00:00<00:00, 20227.55case/s]
{'predictions_path': 'outputs/predictions/vstar_bench_predictions.jsonl', 'candidates_path': 'outputs/predictions/vstar_self_consistency_n4_candidates.jsonl', 'total_predi
ctions': 191, 'bad_cases': 191, 'bad_case_rate': 1.0, 'category_counts': {'parse_failure': 191}, 'problem_type_counts': {'multiple_choice_visual_reasoning': 189, 'counting
': 1, 'chart_or_table': 1}, 'sampled_cases': 8}
```